

Hugo Latourelle-Vigeant

Department of Statistics and Data Science – Yale University
New Haven, CT, USA

✉ hugo.latourelle-vigeant@yale.edu • 🌐 hugol-v.github.io

Education

Ph.D. in Statistics and Data Science

2024–Present (Expected 2030)

Yale University, New Haven, CT

- Advised by Professor Theodor Misiakiewicz

M.Sc. in Mathematics and Statistics

2022–2024

McGill University, Montreal, QC

- Thesis: The matrix Dyson equation for machine learning: Correlated linearizations and the test error in random features regression
- Co-advised by Professor Courtney Paquette and Professor Elliot Paquette

B.Sc. Joint Honours in Mathematics and Computer Science

2018–2022

McGill University, Montreal, QC

Research Interests

Random matrix theory, machine learning theory, statistical-computational trade-offs, and high-dimensional probability.

Publications & Preprints

- [1] **Latourelle-Vigeant, H.**, Chewi, S., Pooladian, A.-A., Sous, J., & Misiakiewicz, T. (2026). “Generalization, memorization, and overfitting for diffusion models trained in the lazy high-dimensional regime.” *arXiv:2608.23938* (Preprint).
- [2] **Latourelle-Vigeant, H.**, & Misiakiewicz, T. (2026). “Statistical-Computational Trade-offs in Learning Multi-Index Models via Harmonic Analysis.” *arXiv:2602.09959* (Preprint).
- [3] **Latourelle-Vigeant, H.**, & Paquette, E. (2026). “Dyson Equation for Correlated Linearizations and Test Error of Random Features Regression.” *Random Matrices: Theory and Applications*, 15(01), 2550026.

Talks & Presentations

- “Generalization, memorization, and overfitting for diffusion models trained in the lazy high-dimensional regime,” seminar talk, *Yale Theory Student Seminar*, Yale University, New Haven, CT, September 2026.
- “Generalization, Memorization, and Overfitting in Kernel Denoising Score Matching in High Dimensions,” poster presentation, *Princeton Machine Learning Theory Summer School*, Princeton University, Princeton, NJ, August 2026.
- “Introduction to the Matrix Dyson Equation,” talk, *ETH Zurich*, March 2024.
- “Matrix Dyson Equation for Correlated Linearizations,” workshop talk, *The Many Facets of Random Matrix Theory workshop*, 2023 Canadian Mathematical Society Winter Meeting,

Montreal, QC, December 2023.

- “**Matrix Dyson Equation for Correlated Linearizations**,” seminar talk, *Seminar in Machine Learning, Optimization and Random Matrix Theory*, McGill University, Montreal, QC, September 2023.
- “**GD and Large Linear Regression: Concentration and Asymptotics for a Spiked Model**,” conference talk, *4th Undergraduate Student Research Conference*, McGill University, Montreal, QC, August 2021.

Teaching Experience

Teaching Fellow for S&DS4100/6100 – Statistical Inference <i>Department of Statistics and Data Science, Yale University</i>	Fall 2026
Teaching Fellow for S&DS2420 – Theory of Statistics <i>Department of Statistics and Data Science, Yale University</i>	Spring 2026
Head Teaching Fellow for S&DS2410 – Probability Theory <i>Department of Statistics and Data Science, Yale University</i>	Fall 2025
Teaching Assistant for MATH 141 – Calculus 2 <i>Department of Mathematics and Statistics, McGill University</i>	F2022, W2023, W2024
Graduate Course Assistant for MATH 463/563 – Convex Optimization <i>Department of Mathematics and Statistics, McGill University</i>	W2023, W2024
Graduate Course Assistant for MATH 417/517 – Linear Optimization <i>Department of Mathematics and Statistics, McGill University</i>	Fall 2022
Undergraduate Course Assistant for MATH 560 – Optimization <i>Department of Mathematics and Statistics, McGill University</i>	Winter 2022

*Note: W = Winter, S = Spring, F = Fall

Academic Service

Organization	
Co-organizer, Montreal RMT-OPT-ML Seminar <i>McGill University</i>	Fall 2023
Review	
Workshop on Principles of Generative Modeling (PriGM) <i>Conference on Neural Information Processing Systems (NeurIPS)</i>	2026
Main Track Reviewer <i>Conference on Neural Information Processing Systems (NeurIPS)</i>	2026
High-dimensional Learning Dynamics (HiLD) Workshop <i>International Conference on Machine Learning (ICML)</i>	2023–2026
OPT Workshop on Optimization for Machine Learning <i>Conference on Neural Information Processing Systems (NeurIPS)</i>	2022–2026

Volunteering.....
OPT Workshop on Optimization for Machine Learning 2020–2021
Conference on Neural Information Processing Systems (NeurIPS)

Work Experience

Data Science Intern Summer 2024
Caisse de dépôt et placement du Québec (CDPQ), Montreal, QC
○ Part of the natural language processing (NLP) team
○ Conducted topic modeling and sentiment analysis on large-scale financial documents

Awards and Scholarships

First-Class Honours in Mathematics and Computer Science 2022
McGill University
Undergraduate Student Research Award (USRA) & FRQNT Supplement 2021
NSERC & FRQNT
Major Entrance Scholarship in Science 2018
Hydro-Québec